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Cardano Governance

Metrics Manual

Commissioned by Cardano DReps; delivered by IO and Cardano Community Members including:

- Fanny Wijaya
- Ken Erik Oelmheim
- Maureen Wepngong (Giiyo Tech)
- Max van Rossem
- Nana Safo (Wada Global)
- Richmond Oppong (Wada Global)
- Tevo Kask

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1. Purpose

This manual provides a set of data sources with step-by-step guidance to calculate each metric referenced in the Governance Measurement Framework. This report is intended for tooling providers to generate consistent measurements, or for technical community members to independently verify a metric.

This metrics manual was first written by the researchers based on the source documents. It was further developed using a structured application of AI tools Gemini CLI and Claude Code. For this manual, AI was primarily used to enforce consistency between the research report and associated metric definitions. All portions of the document that were built using AI tools have been reviewed by the researchers to ensure efficacy and alignment with the goals of the proposed measurement methodology.

2. Data Sources

CARDANO NODES

A running cardano-node that validates and relays blocks and transactions, maintains the canonical ledger state locally and serves as the authoritative source of raw on-chain events (blocks, transactions, epochs, stake, registration). Often paired with cardano-db-sync for structured querying.

QUALITATIVE DATA

The measurement framework collects both on-chain quantitative data and off-chain qualitative data. Qualitative data collection utilizes surveys, interviews, and workshops to access insights that would otherwise not be possible to measure using on-chain data alone - the "why" corresponding to the "what" measured by the "how" on-chain metrics. Qualitative data gathering also uncovers feedback related to elements of Cardano's governance that are not on-chain, such as the need for new governance features, off-chain consensus processes, impacts of influencers, or tooling designs on voting outcomes. These qualitative data collection tools are constructed from questions that directly tie to on-chain measurement data points, plus a set of open-ended questions with the goal of creating actionable insights corresponding to both the on-chain data and what cannot be measured on-chain, supporting process and resource improvements.

3. Measurement Logic

3.1. ADA OWNER METRICS

3.1.1. PERCENTAGE OF CIRCULATING ADA ACTIVELY VOTING ON GOVERNANCE (EXCLUDING ABSTAIN)

Definition: The percentage of all ada in circulation that actively participates in on-chain governance by casting a Yes, No and no-confidence vote. This shows active participation by measuring the part of ada that is involved in making decisions.

Step-by-Step Logic of How It's Calculated:

1. Find the snapshot epoch: Use the ledger state at the most recent epoch boundary (the one right before votes are counted), as CIP-1694 requires using the latest per-DRep stake distribution[1].
2. Find out how much ada is in circulation: Calculate the total amount of ada that is in circulation at that time (the total amount of ada that is not in the treasury, which is usually the sum of all UTxO values).
3. Calculate active voting ada: Query the on-chain delegation state (via cardano-db-sync). Sum the stakes of all stake credentials that are given to a registered DRep or the "No Confidence" option. CIP-1694 says that ada delegated to a DRep is an active voting stake[1], and "No Confidence" delegations are counted as votes[1].
4. Subtract abstentions: Sum the ada that was given to the always abstain option.
   CIP-1694 makes it clear that any stake given to always abstain is not part of the active voting stake[1].
5. Compute percentage: To find the percentage, take the total active stake and subtract the abstained stake. Then divide that number by the total amount of ada that is in circulation. To get a percentage, multiply by 100. For example: *if 20 billion ada is in circulation and 5 billion is actively delegated with 0.5 billion delegated to always Abstain, the metric is* $(5 - 0.5) / 20 \times 100 = 22.5\%$.

Data Presentation/Visualization: Time series line chart tracking the percentage across epochs with a horizontal reference line marking the 50% legitimacy threshold established by the Guardrails appendix in the Cardano Constitution.

Why This Logic Is Correct: CIP-1694's rules say exactly which ada are "active voting stake." The calculation uses the on-chain snapshot and delegation certificates to make these rules exactly the same. Using the previous-epoch snapshot for DRep delegations makes sure that the timing is right based on CIP-1694. This is because it

always uses the most recent per-DRep stake distribution at the epoch boundary[1]. The CIP says that you can't include Abstain-delegated stake[1].

3.1.2. NUMBER OF UNIQUE STAKE ADDRESSES DELEGATED TO DREPS

Definition: The number of different stake addresses (also known as staking keys) that have given their voting rights to a registered Delegated Representative (DRep) and script DRep. This metric only looks at addresses that are giving a script / DRep their trust and not those that are choosing the fixed always abstain options.

Step-by-Step Logic of How It's Calculated:

1. Select snapshot: Use the stake delegation snapshot from the last epoch boundary (per CIP-1694, which uses current stake distribution[1]).
2. Query delegations: In the db-sync database, look at all stake delegation certificates. Each of these certificates connects one stake credential to a DRep. Each certificate that does not return null (Not delegated to a script / DRep) or Always abstain will be counted.
3. Count unique addresses: Get the set of unique stake credential hashes (or stake address IDs) from the delegations that you filtered.
4. Aggregate: This set's size is the metric. Use SQL, then do SELECT COUNT(DISTINCT stake_address_id) FROM stake_delegation WHERE delegate = 'DRep_id'.
   Note: Delegations can change over time.

Data Presentation/Visualization: Time series line chart showing cumulative growth of unique stake addresses over epochs to track broadening participation.

Why This Logic Is Correct: By directly counting on-chain delegations to script / DReps, the logic exactly measures what CIP-1694 says is the way for ada owners to take part[1]. It follows the protocol's own rules for registering and delegating: an address becomes part of the voting process by delegating to a DRep. Using the official snapshot makes sure that the results are correct and can be repeated. Since every delegation and DRep registration is recorded on-chain, running the same query on db-sync yields an identical count each time.

3.1.3. STICKINESS OF ADA OWNER TO DREP DELEGATION

Definition: The rate at which stakeholders move their delegation from one script / DRep to another over time. This is measured by the percentage of delegations that change from one epoch to the next. It shows how easily ada owners reassign their voting power among DReps and script DReps.

Step-by-Step Logic of How It's Calculated:

1. Choose time window: Choose a time frame for measuring changes, like every epoch. Use delegation snapshots at the beginning and end of that period.
2. Gather delegation states: Using db-sync, get the stake credential → DRep / script certificate mapping at the start (epoch E_0) and at the end (epoch E_1).
3. Compare delegations: For each stake address, check to see if the delegated script / DRep ID has changed between E_0 and E_1. If an address's delegate at E_0 is different from its delegate at E_1, count it as "changed." This does not include cases where it was undelegated or always abstained.
4. Find the frequency: The metric can be expressed as (number of changed addresses) / (total addresses delegated to DReps)x 100% for that period. One can average changes per epoch for ongoing monitoring.

Data Presentation/Visualization: Time series line chart showing the percentage of stake addresses that changed their DRep delegation per epoch, smoothed with a moving average.

Why This Logic Is Correct: This approach directly implements the definition of "changing DRep." It relies on on-chain delegation certificates: every time an ada owner re-delegates, a new certificate appears in the ledger. By comparing snapshots, we capture exactly those updates.

3.2. DREP METRICS

3.2.1. GINI COEFFICIENT OF DELEGATED VOTING POWER AMONG SCRIPT / DREPS

Definition: A statistical measure between 0 and 1 that quantifies the inequality of voting power distribution among active DReps and script DReps. It is calculated based on the total ada stake delegated to each entity. A value of 0 indicates perfect equality (all DReps have the same amount of stake), while a value of 1 indicates perfect inequality (one DRep controls all delegated stake). A lower Gini coefficient signifies a more decentralized and evenly distributed voting power, implying greater resistance to centralized control.

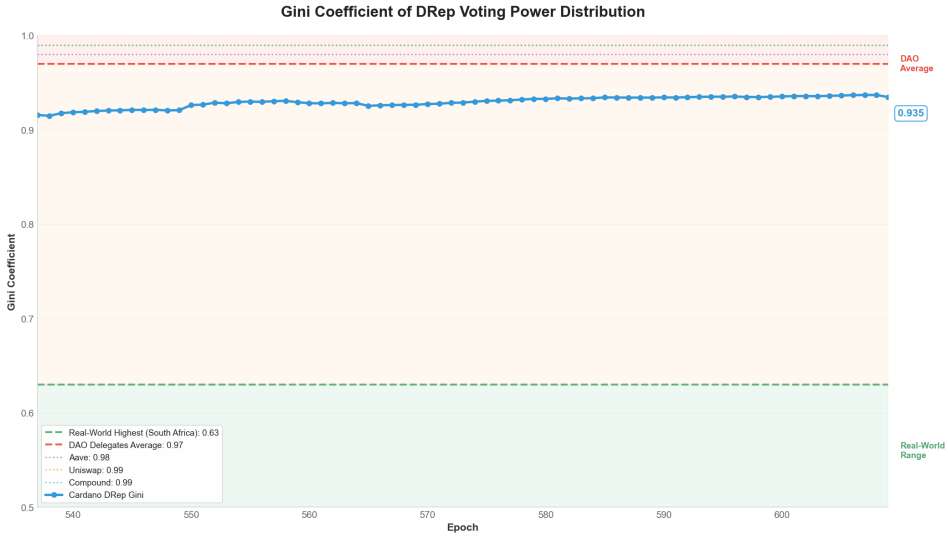
Step-by-Step Logic of How It's Calculated:

1. Snapshot stake distribution: Use the epoch-boundary stake snapshot for DReps and script DReps per CIP-1694.[1].
2. Aggregate by DRep: Sum all Ada delegated to active script / DReps, excluding "Abstain" delegations.
3. List of weights: Create a vector $x_1, x_2, \ldots, x_N$ representing the total delegated stake (in Lovelace) for each active script or DRep ID.

4. Compute Gini: Apply the standard Gini coefficient formula to the vector of stake totals.

$$G = \frac{2\sum_{i=1}^{n} i x_i}{n\sum_{i=1}^{n} x_i} - 1 - \frac{1}{n}$$

This yields a value in[0,1]. Standard data analysis tools and libraries can compute this directly from the vector.



Data Presentation/Visualization:

- Type: Single-line time series chart with reference thresholds.
  - X-Axis: Epoch number.
  - Y-Axis: Gini Coefficient (Number between 0 and 1).
- Data Line: Solid orange line representing the Gini Coefficient.
- Reference Lines:
  - Perfect Equality (0): Dashed gray line (optional, for baseline).
  - Perfect Inequality (1): Dashed gray line (optional, for baseline).

Why This Logic Is Correct: The calculation uses exactly the on-chain, epoch-based distribution that CIP-1694 defines as DRep voting power. Because DRep voting weight is literally the ada delegated to them, summing and comparing these totals captures inequality precisely and transparently. Using a standard Gini formula is appropriate: it is a well-known, reproducible statistic for distribution inequality.

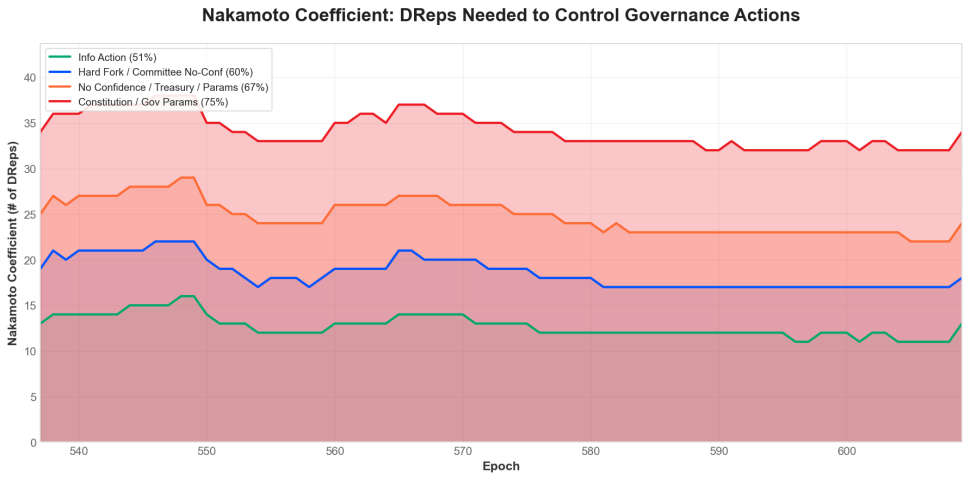
3.2.2. NAKAMOTO COEFFICIENT FOR DREP POWER

Definition: An integer count representing the minimum number of DReps and Script DReps whose combined delegated voting power exceeds the required majority threshold for specific governance actions. It is calculated by ranking all active DReps

by their delegated stake and counting how many are needed until the cumulative sum crosses a defined threshold. A higher value indicates stronger resistance to collusion and governance capture.

Step-by-Step Logic of How It's Calculated:

1. Snapshot stake distribution: Use the epoch-boundary stake snapshot for DReps and script DReps per CIP-1694.[1].
2. Aggregate active voting stake: Sum all Ada delegated to active script / DReps, excluding "Abstain" delegations.
3. Ranks DReps by delegated power: Sort all active DReps in descending order based on their delegated stake totals.
4. Cumulative summation for multiple thresholds: Sequential summation is performed until the cumulative total first exceeds distinct constitutional threshold:
    - 51% (Info actions)
    - 60% (Hard Fork / Committee No-Confidence)
    - 67% (No Confidence / Treasury / Protocol Parameters)
    - 75% (Constitution / Governance Parameters)
5. Count = Nakamoto Coefficient: The number of DReps required to reach each respective threshold is recorded per epoch.



Data Presentation/Visualization:

- Type: Multi-line time series chart with shaded area regions.
    - X-Axis: Epoch number.
    - Y-Axis: Nakamoto Coefficient (Number of DReps).
- Threshold Lines:
    - Info Action (51%): Solid green line.
    - Hard Fork / Committee No-Conf (60%): Solid blue line.
    - No Confidence / Treasury / Params (67%): Solid orange line.
    - Constitution / Gov Params (75%): Solid red line.

- Shading: Transparent color-coded fills between the lines to visually distinguish the decentralization requirements for different governance action types.

Why This Logic Is Correct: While the standard Nakamoto Coefficient (Minimum Attack Vector) typically uses a 51% majority, this implementation tracks multiple thresholds defined by the Cardano Constitution and governance parameters. This provides a more granular view of "collusion resistance" across different action types. By mapping these specific thresholds on a time series, stakeholders can monitor the evolving coordination requirements for everything from simple info actions to high-stakes constitutional changes.

3.2.3. PERCENTAGE OF DREPS VOTES THAT INCLUDE A PUBLISHED RATIONALE (ANCHOR URL + HASH)

Definition: The fraction of governance votes cast by DReps and script DReps that attach an external rationale document via an on-chain anchor (a URL plus its content hash). Because of the nature of the on-chain data, rationales that are low quality or not informative cannot be identified without human review. Because of the nature of the on-chain data process this evaluation is out of scope for this framework.

Step-by-Step Logic of How It's Calculated:

1. Collect Drep votes: Use db-sync to get all vote transactions for a certain epoch, where the role is DRep. There is an optional anchor field (URL and hash) in each vote record.
2. Identify anchors: For each DRep and script DRep vote, make sure that the anchor fields are not empty (they should have a URL and a hash). If there is an anchor, count that vote as "with rationale."
3. Compute percentage: To find the percentage, use Dtotal = total DRep votes cast in the period and rationale = count with anchors. The formula for the metric is Drationale / Dtotal × 100%.

Data Presentation/Visualization: Time series line chart tracking the percentage of DRep votes with published rationale over epochs.

Why This Logic Is Correct: This logic is directly in line with the rules for governance: the CIP says that you can attach anchors to votes, and the community guidelines say that you should [1][5].

3.2.4. NET CHANGE IN ACTIVE DREP COUNT (REGISTRATIONS - DE-REGISTRATIONS - INACTIVITY)

Definition: The rate of change in the number of active DReps over a standardized measurement period. This metric tracks the health of the DRep population by measuring how quickly the number of active DReps is increasing or decreasing. A

positive rate indicates growing decentralization and representation, while a negative rate may indicate diminishing interest. "Active" means a DRep is registered, not voluntarily retired, and has voted within the drepActivity period (currently 20 epochs). The standardized measurement window is set at 12 epochs to reconcile governance action expiration (6 epochs) and DRep inactivity timeout (20 epochs).

Step-by-Step Logic of How It's Calculated:

1. Select measurement period: Use a standardized window of 12 epochs for measurement. This period balances governance action lifecycles (6 epoch expiration) with DRep activity requirements (20 epoch inactivity timeout).
2. Count new Drep registrations (R): Query cardano-db-sync for all DRep registration certificates issued during the 12-epoch measurement period. Each registration certificate creates a new DRep credential.
3. Count voluntary retirements (D): Query cardano-db-sync for all DRep retirement certificates issued during the 12-epoch measurement period. These represent DReps who voluntarily deregistered.
4. Count DReps marked inactive (I): Identify DReps marked inactive by the protocol during the measurement period per the drepActivity parameter. CIP-1694 specifies that DReps who fail to vote for 20 consecutive epochs are automatically marked inactive and removed from active voting stake calculations.
5. Calculate net change rate: Apply the formula: Net change rate = [R - (D + I)] / 12 epochs. This yields a rate per epoch showing how quickly the DRep population is growing or declining. A positive value indicates net growth; a negative value indicates net decline.

Data Presentation/Visualization: Waterfall chart displaying the components driving net change: new registrations (positive, green), voluntary retirements (negative, red), and inactivity removals (negative, orange), with the net rate per epoch displayed prominently.

Why This Logic Is Correct: The calculation uses exact on-chain mechanisms defined in CIP-1694 for the DRep lifecycle. DRep registration and retirement are recorded via explicit certificates on-chain, while the protocol automatically enforces inactivity marking through the drepActivity parameter (currently 20 epochs). Dividing by the measurement period (12 epochs) produces a rate of change that enables comparison across different time periods and tracking of trends. The 12-epoch window is chosen to balance governance action lifecycles with DRep activity requirements, providing a meaningful measurement interval that captures DRep population dynamics without being too short (volatile) or too long (lagging).

3.3. SPO METRICS

3.3.1. SPO ENGAGEMENT IN GOVERNANCE ACTIONS – OVERALL AND MANDATORY ONLY

Definition: The extent to which Stake Pool Operators (SPOs) participate in on-chain governance votes. This metric measures SPO engagement through two separate percentages: (1) Overall engagement - the percentage of SPOs (by voting power) who vote on each governance action regardless of type, and (2) Mandatory-only engagement - the percentage of SPOs (by voting power) who vote specifically on mandatory governance actions that require SPO approval (Motion of No-Confidence, New Constitutional Committee, Update to the Constitution, Hard-Fork Initiation, and certain Protocol Parameter changes). Overall participation is a core metric of SPO engagement, while low participation from SPOs in mandatory actions could compromise security and network stability per the Constitution's security principle. When calculating percentages, any stake that was assigned to the always abstain option is treated as non-participation.

Step-by-Step Logic of How It's Calculated:

1. Identify all governance actions and mandatory subset: Query db-sync to extract all governance actions submitted during the measurement period. From this set, filter for mandatory actions requiring SPO approval according to CIP-1694 (Motion of No-Confidence, New Constitutional Committee, Update to the Constitution, Hard-Fork Initiation, and security-relevant Protocol Parameter changes). See the Governance actions spec for the complete list of action types requiring SPO approval.
2. Get pool parameters and stake identities: Query db-sync to extract Stake Pool Registration Certificates, retrieving each Pool_ID and its associated active_stake at the relevant epoch boundary. This mapping identifies which stake belongs to which SPO.
3. Compute delegated stake per SPO: For each Pool_ID, sum the total delegated stake assigned to that pool. This represents the SPO's voting power.
4. Subtract always abstain stake: Query how much of the stake delegated to each pool is assigned to the always Abstain delegation option. Subtract that abstained amount from each pool's total delegated stake to calculate the effective voting stake per SPO.
5. Extract SPO votes per action: For each governance action (both overall and mandatory subset), use db-sync to identify which Pool_IDs cast votes (Yes, No, or Abstain on the action itself). Track the voting power behind each vote.
6. Calculate Overall Engagement: For each governance action in the measurement period:

- Sum the voting power of all SPOs who voted (Yes, No, or Abstain) on that action
- Sum the total SPO voting power (all active SPOs' delegated stake minus always abstain stake)
- Calculate: (Voting SPO power) / (Total SPO power minus always abstain) × 100%
- This gives the percentage of SPO voting power that participated in each action

7. Calculate Mandatory-Only Engagement: Apply the same calculation as step 6, but filter to include only mandatory actions:
    - For each mandatory action, calculate: (Voting SPO power on mandatory action) / (Total SPO power minus always abstain) × 100%
    - This gives the percentage of SPO voting power that participated specifically in mandatory governance actions
8. Aggregate across actions: Calculate average engagement rates across all actions (for overall metric) and across all mandatory actions (for mandatory-only metric). Track trends over time by epoch.

Data Presentation/Visualization: Time series line chart tracking SPO engagement percentage over epochs with two lines: one for overall SPO engagement across all governance actions and one for mandatory-only SPO engagement. Include a horizontal reference line at 50% representing the legitimacy threshold from the Constitution's Guardrails appendix.

Why This Logic Is Correct: The calculation uses the on-chain vote records for SPOs on each action, which are maintained by the ledger. CIP-1694's vote model (stake-weighted voting) is used directly – SPO voting weight is proportional to delegated stake [3]. By measuring both overall engagement and mandatory-specific engagement, we capture the full picture of SPO fulfillment of their governance role: general participation in governance processes (overall) and critical security oversight (mandatory). Treating always abstain stake as non-participation aligns with CIP-1694's definition that abstain-delegated stake is excluded from active voting stake calculations [1].

3.3.2. STICKINESS OF ADA DELEGATED TO SPOS (STAKE THAT REMAINS OVER MULTIPLE EPOCHS)

Definition: The amount or percentage of ada stake delegated to each stake pool that has remained with the same pool across multiple epochs. In other words, the quantity of stake that does not redelegate away from a pool between successive snapshots. This is sometimes called "loyal" or "sticky" stake.

Step-by-Step Logic of How It's Calculated:

1. Select epochs: Select two snapshots that are a set amount of time apart (for example, 6 epochs). These will be the epochs E_0 and E_1. The earliest epoch you can use is always 2 epoch boundaries back from the current epoch.
2. Get delegation states: Use db-sync to write down the pool ID that each stake address is delegated to at epoch E_0 and E_1.
3. Identify sticky stakes: For each stake address that was delegated at E_0, check if its delegation target at E_1 is the same pool. Sum the ada for all of the addresses whose pool ID stayed the same.
4. Compute metric: Divide the sticky stake sum by the total stake given at E_0 Data Presentation / Visualization: Time series line chart showing the percentage of sticky stake (remaining with same pool) measured across different time intervals (3 epochs, 6 epochs, 12 epochs).

Data Presentation/Visualization: Time series line chart showing the percentage of sticky stake (remaining with same pool) measured across different time intervals (3 epochs, 6 epochs, 12 epochs).

Why This Logic Is Correct: The method uses on-chain delegation snapshots, so it captures actual delegation behavior. Delegation certificates are recorded in each epoch, and comparing them across epochs precisely identifies stake addresses that did not move to a new SPO.

3.4. CC METRICS

3.4.1. AVERAGE CONSTITUTIONAL COMMITTEE RESPONSE TIME TO GOVERNANCE ACTIONS

Definition: The average time (measurement in slot's) between when a governance action is submitted on-chain and when Constitutional Committee (CC) members cast their votes at governance. This measures the efficiency of the constitutional review process and identifies potential bottlenecks that could delay governance system operation.

Step-by-Step Logic of How It's Calculated:

1. Identify eligible governance actions: Query all on-chain governance actions submitted within the measurement period from cardano-db-sync. Exclude "Motion of no-confidence" or "Update Constitutional Committee" and Update committee/threshold (state of no-confidence) actions targeting the CC itself.
2. Record submission and vote timestamps: If a CC member does not vote within 6 epochs, that CC member will get 2.590.000 slots added. For each governance action, extract the submission timestamp from the governance action certificate. Query all CC vote certificates for that action and record

each CC member's vote timestamp (Slot number) If no vote has been done add 2.590.000 slots to each CC.

3. Calculate response time per action: For each governance action, compute the time elapsed from submission to each CC member vote. Use the median CC member vote timestamp as the representative response time for that action.
4. Aggregate across actions: Calculate median response time, mean response time, and interquartile range across all governance actions. Segment by governance action type (Treasury, all Protocol Parameters, Hard Forks, Update Constitution) to identify whether critical actions receive faster review.

Data Presentation/Visualization: Time series line chart tracking median CC response time (in days) across epochs, with separate colored lines for different governance action types.

Why This Logic Is Correct: CIP-1694 establishes that governance actions are "applied on an epoch boundary"(Ratified or enacted) [2] and must be ratified by governance bodies including the CC. The Constitution mandates that "no governance action, other than a 'Motion of no confidence,' or 'Update Constitutional Committee/threshold and/or term' may be implemented on-chain unless a requisite percentage of the members of the Constitutional Committee. shall have first determined and affirmed through an on-chain action that such proposal does not violate this Constitution" [5]. Using median vote timestamp accounts for independent CC member voting.

3.4.2. PERCENTAGE OF CONSTITUTIONAL COMMITTEE VOTES INCLUDING RATIONALE

Definition: The percentage of Constitutional Committee votes that include published rationale (via anchor URL and hash) explaining the constitutional basis for the decision. This measures transparency and accountability in constitutional review, particularly compliance with the Constitution's mandate to explain unconstitutional determinations.
Metadata is represented as in CIP-0136.

Step-by-Step Logic of How It's Calculated:

1. Query CC votes: Use cardano-db-sync to extract all CC vote certificates for the measurement period. Each vote record includes the governance action ID, CC member ID, vote choice (Yes/No/Abstain), timestamp, and optionally an anchor (URL + hash).
2. Check for anchor presence and validity: For each vote, determine whether an anchor is attached to the vote certificate. If present, retrieve the content

from the URL and verify the hash matches the on-chain hash (integrity check).

3. Calculate publication rate: Count votes with valid anchors and divide by total CC votes cast: (Votes with valid anchors) / (Total votes) × 100%. Segment by vote decision: calculate separately for No votes (unconstitutional) versus Yes votes (constitutional).
4. Segment by member and action type: Calculate individual CC member publication rates and segment by governance action type to identify patterns in transparency compliance.

Data Presentation/Visualization: Stacked bar chart showing percentage of CC votes with published rationale versus without rationale for each epoch.

Why This Logic Is Correct: CIP-1694 specifies that votes include "an optional anchor" which is "a pair of: a URL to a JSON payload of metadata(CIP-0136), a hash of the contents of the metadata URL"[4]. The Constitution explicitly mandates: "When voting that a governance action proposed to be executed on-chain is unconstitutional, the Constitutional Committee collectively, or each member of the Constitutional Committee casting such a vote separately, shall set forth the basis for its decision with reference to specific Articles of this Constitution"[5]. Tracking No vote publication rates verifies compliance with this requirement.

3.4.3. PERCENTAGE OF CONSTITUTIONAL COMMITTEE ABSTAIN RATE

Definition: The percentage of Constitutional Committee votes cast as Abstain (rather than Yes or No). This measures the CC's ability to reach definitive constitutional determinations, with high abstention potentially indicating constitutional ambiguity, conflicts of interest, or expertise gaps.

Step-by-Step Logic of How It's Calculated:

1. Query CC votes: Use cardano-db-sync to extract all CC vote certificates for the measurement period. For each vote, identify the vote choice field: Yes (constitutional), No (unconstitutional), or Abstain.
2. Filter for actions not allowing for CC review: No-confidence and update CC actions targeting the CC itself. Focus on the five action types requiring constitutional review: Treasury, Protocol Parameters, Hard Forks, New Constitution and Info action.
3. Calculate abstention rate: Count total Abstain votes and divide by total CC votes on eligible actions: (Abstain votes) / (Total votes on eligible actions) × 100%.

4. Segment by member and action type: Calculate abstention rates separately for each governance action type and for each CC member to identify systematic patterns.

Data Presentation/Visualization: Time series line chart tracking CC abstention rate across epochs, with separate colored lines for different governance action types to identify which proposal categories generate highest uncertainty.

Why This Logic Is Correct: CIP-1694 establishes three voting options including Abstain [2]. The Constitution mandates that the CC "shall have first determined and affirmed through an on-chain action that such proposal does not violate this Constitution" [5], implying reaching a Yes/No determination. The Constitution requires the CC to be "free and independent in making decisions" [5], and high abstention rates undermine this constitutional gatekeeping function by preventing definitive review.

4. Frequency of Data Collection

4.1. QUANTITATIVE ON-CHAIN DATA

Although the data in this report was collected at one point in time, this data collection can be automated, with frequent measurement and re-measurement feasible. Many metrics specify using the ledger state at the most recent epoch boundary to ensure consistent data, and several use a standardized window of measurement, such as 6 (Stickiness of ada delegated to SPOs) or 12 (Net change in active DRep count) epochs, or slot-based measurement (Average Constitutional Committee Response Time to Governance Actions). In other cases, discretion regarding measurement frequency is given to the analysis and can be chosen based on context. Precise details about how frequently to gather a datapoint from Cardano are specified within the metrics definition and manual.

4.2. QUALITATIVE OFF-CHAIN DATA

Qualitative feedback is resource intensive. Collection requires members of the ecosystem to spend time providing their feedback, and it requires human analysis of the results, even when partially automated through large language model analysis. The minimum recommendation is to use quantitative triggers for emergency action - metrics reaching a dangerous value or trend triggers an emergency response protocol, which would be an extension of the technical protocols outlined in CIP-135. These protocols may require qualitative data gathering to fully resolve.

Beyond establishing emergency protocols, the frequency of qualitative data gathering will be subject to community interest and DRep funding approval. Sustainable, resource-efficient approaches include:

1. Establish a periodic checkpoint for a holistic review. This should be on the scale of 3-5 years and triggered when on-chain metrics are characteristically different than the previous checkpoint.
2. Between periodic checkpoints, make retrospectives a norm for governance processes: Manual retrospectives and pop-up questions in governance tools require marginal effort for participants compared to an isolated set of governance surveys. Feedback will be based on fresh recollections rather than faded memories.
3. Enable retrospectives at scale. Create a single place to host a standard set of governance questions and central repository for responses. Distributing collection across existing governance engagement points reduces survey fatigue by embedding questions into natural workflows rather than requiring separate research participation. This approach also unlocks insights across various institutions, governance tools and retrospectives. This could be automated via APIs.
4. Integrate the governance metrics into the 2030 Cardano Vision KPIs and tracking processes that develop around the KPIs.

In general, when collecting qualitative data, look for opportunities to collect a representative volume, collect information when it's fresh, and piggyback off of existing stakeholder engagement.

5. Bibliography

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